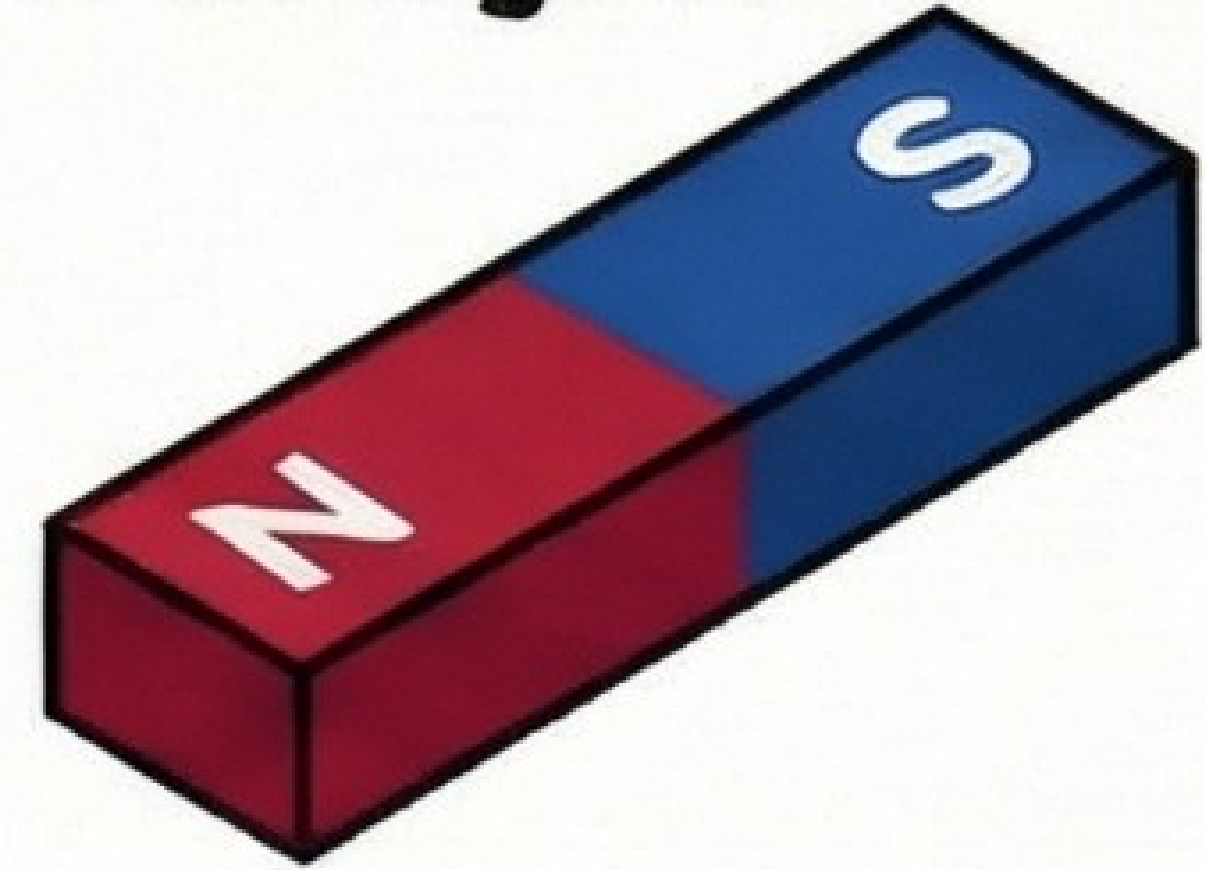


Magnetism & Matter: A Visual Revision Guide

1. The Fundamentals of a Magnet



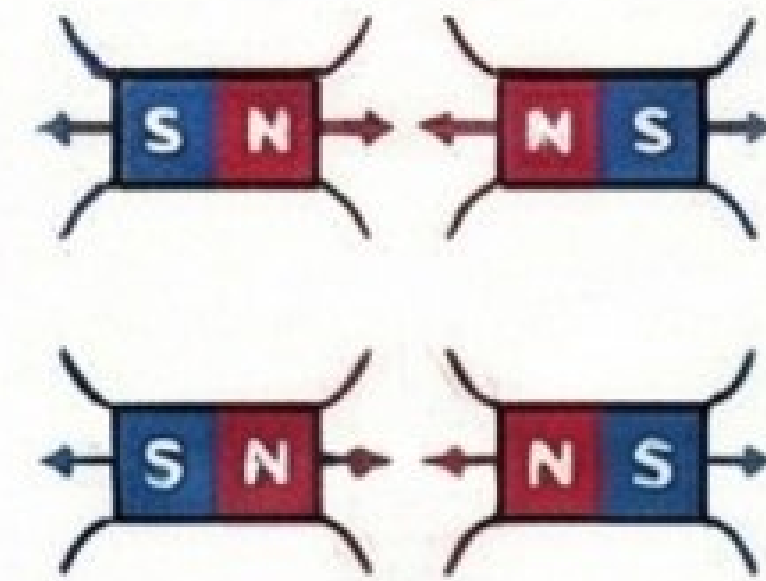
What is a Magnet?

Any substance that can create its own magnetic field and exert a force on other magnetic materials like iron and steel.

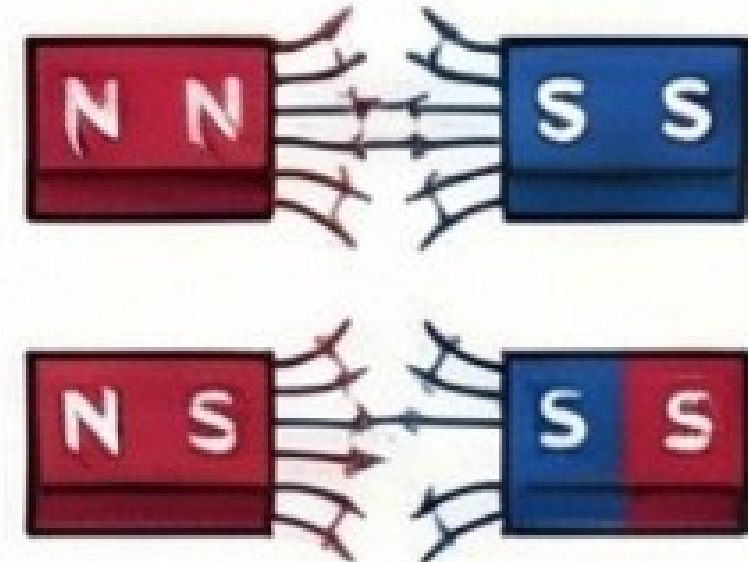
Directive Property



Pole Interaction



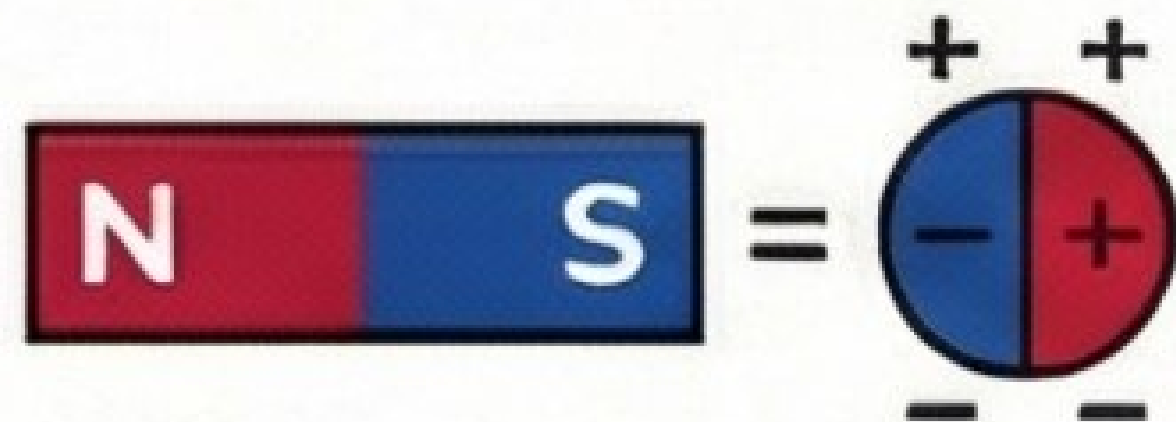
Pole Interaction



Monopoles Do Not Exist



2. The Bar Magnet as a Dipole



An Analogy to the Electric Dipole

North Pole is analogous to a positive charge, and the South Pole to a negative charge.

Magnetic Dipole Moment



$M = m \times 2l$
(where 'm' is pole strength and '2l' is magnetic length).

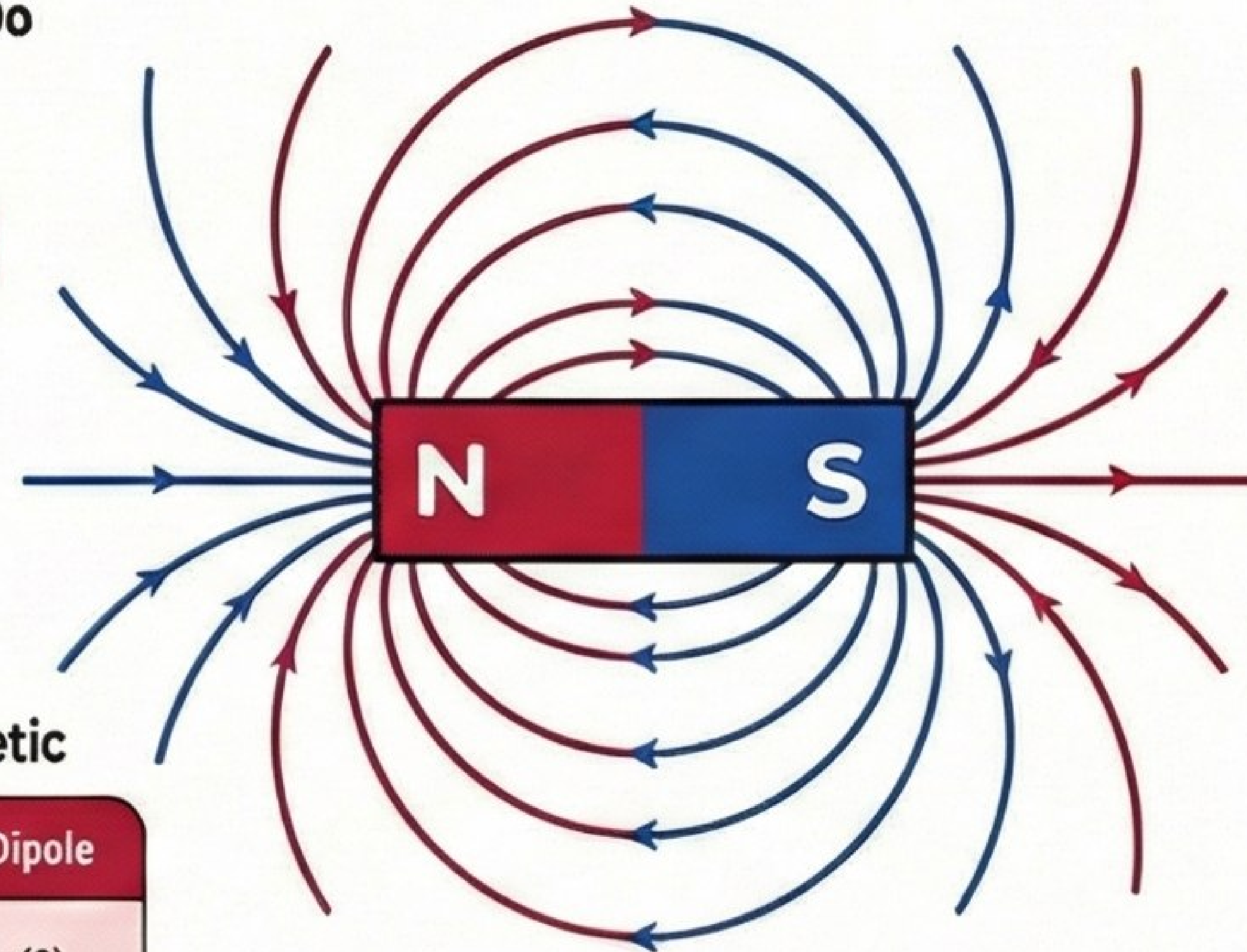
Key Formulas: Electric vs. Magnetic

Concept	Electric Dipole	Magnetic Dipole
Torque	$\tau = pE \sin(\theta)$	$\tau = MB \sin(\theta)$
Potential Energy	$U = -pE \cos(\theta)$	$U = -MB \cos(\theta)$
Work Done (Rotation)	$W = pE(\cos\theta_1 - \cos\theta_2)$	$W = MB(\cos\theta_1 - \cos\theta_2)$
Axial Field	$E = k \frac{2p}{r^3}$	$B = \frac{\mu_0}{4\pi} \frac{2M}{r^3}$
Equatorial Field	$E = k \frac{p}{r^3}$	$B = \frac{\mu_0}{4\pi} \frac{M}{r^3}$

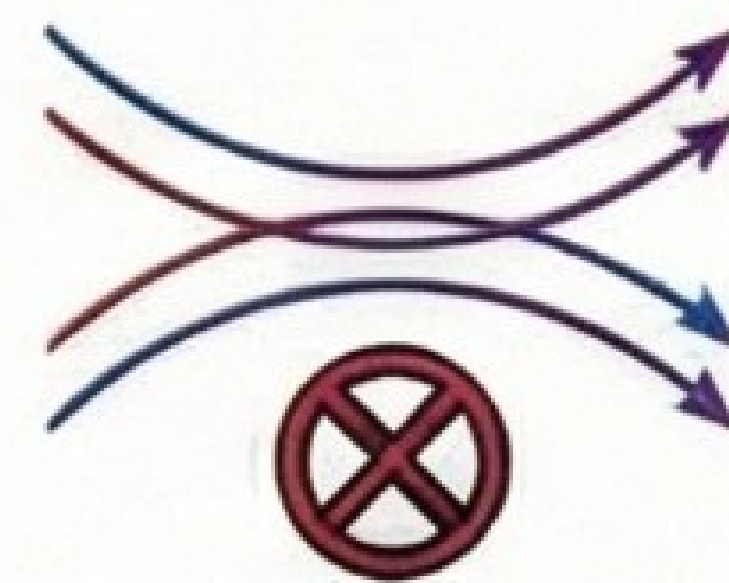
3. Magnetic Fields & Gauss's Law

Magnetic Field Lines Form Closed Loops

They travel from N to S outside and S to N inside, completing a full loop. They never terminate.

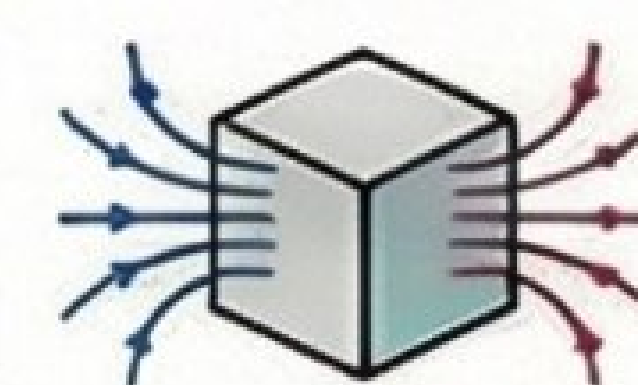


Field Lines Never Intersect



If they did, it would mean two possible directions for the magnetic force at a single point, which is impossible.

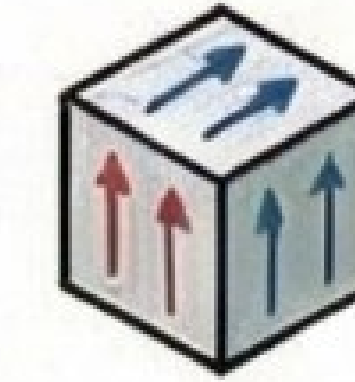
Gauss's Law for Magnetism



Gauss's Law for Magnetism

The net magnetic flux through any closed surface is always zero. For every field line exiting, another must enter.

4. How Materials Respond to Magnetism



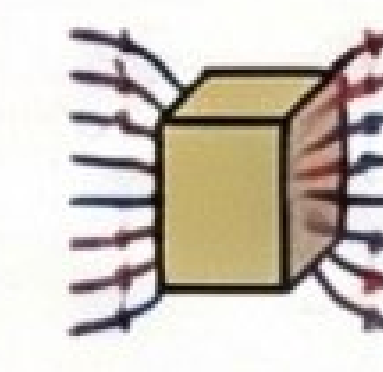
Intensity of Magnetization (I or M)
Net magnetic dipole moment developed per unit volume of a material in a magnetic field.
Unit: A/m.



Magnetizing Intensity (H)
Strength of external magnetic field's ability to magnetize a material.
Unit: A/m.



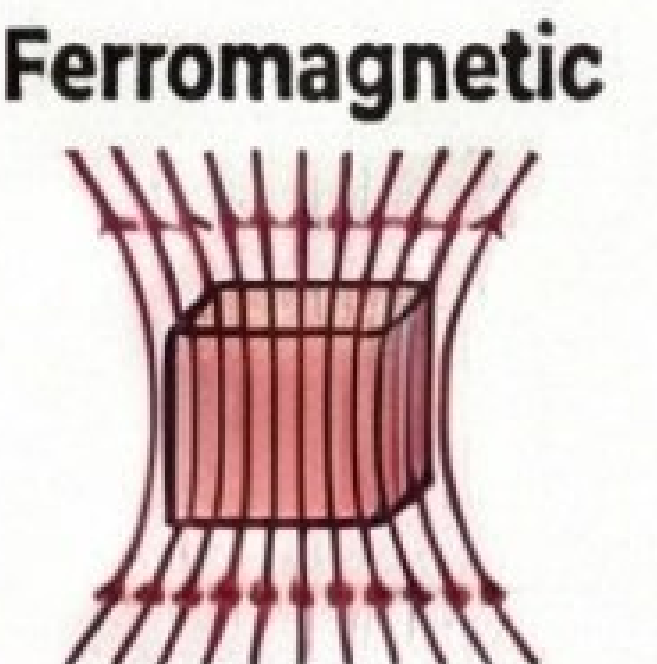
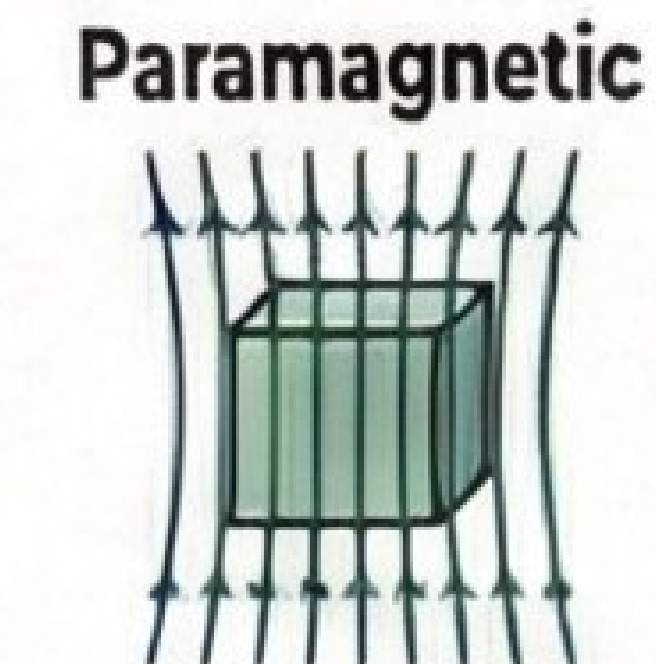
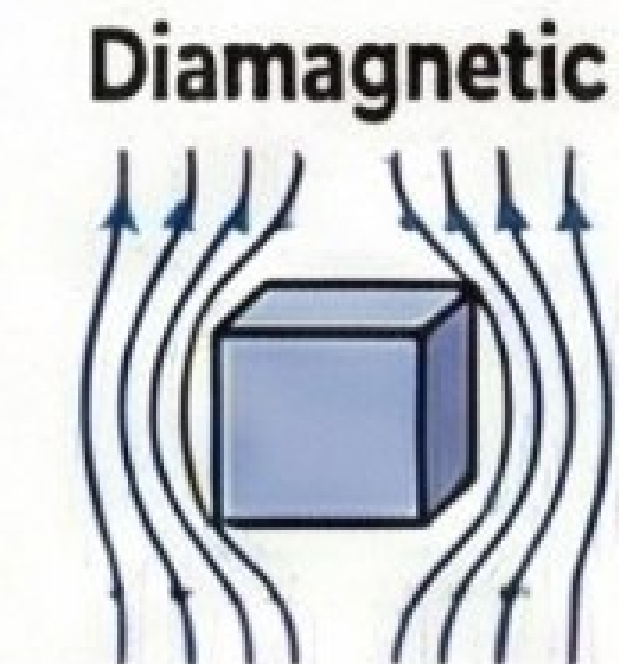
Magnetic Susceptibility (χ)
Measure of how easily a material can be magnetized.
 $\chi = I / H$.
Unitless.



Relative Permeability (μ_r)
Compares material's permeability to free space. Indicates ability to concentrate field lines.
 $\mu_r = 1 + \chi$. Unitless.

5. Classification of Magnetic Materials

How Field Lines Pass Through Materials



Property	Diamagnetic	Paramagnetic	Ferromagnetic
Interaction	Weakly repelled	Weakly attracted	Strongly attracted
Examples	Copper, Gold, Bismuth	Aluminum, Calcium, Oxygen	Iron, Nickel, Cobalt, Steel
Susceptibility (χ)	Small & Negative	Small & Positive	Large & Positive (>1)
Rel. Permeability (μ_r)	Slightly less than 1 ($0 < \mu_r < 1$)	Slightly greater than 1	Much greater than 1 ($\mu_r \gg 1$)
Effect of Temperature	Independent	Inversely proportional	Decreases, vanishes above Curie Temperature
Motion in Field	Moves from stronger to weaker parts	Moves from weaker to stronger parts	Moves rapidly from weaker to stronger parts